RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



Python Programming for Design

Laboratory Record

Name : Pooja K

Year / Branch / Section : 2nd Year / Computer Science and Design / None

Register No. : 2116251701039

Semester : 3rd Semester

Academic Year : 2026 – 2027

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM – 602 105.

BONAFIDE CERTIFICATE

NAME Pooja K

ACADEMIC YEAR 2026 – 2027 **SEMESTER** 3rd **BRANCH** Computer Science and Design

UNIVERSITY REGISTER NO. 2116251701039

Certified that this is the bonafide work done by the above student in the ……Python Programming for Design. Laboratory during the year 2026 – 2027.

Signature of Faculty – in – Charge

Submitted for the Practical Examination held on ……

Internal Examiner **External Examiner**

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Reg. No. : 2116251701039 Name : Pooja K

Year : 2nd Branch : Computer Science and Design Sec : None

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Started on	Thursday, 23 July 2026, 4:05 PM
State	Finished
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Time taken	22 hours 24 mins
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 10.00 out of 10.00

Rohit wants to add the last digits of two given numbers.

For example,

If the given numbers are 267 and 154, the output should be 11.

Below is the explanation:

Last digit of the 267 is 7

Last digit of the 154 is 4

Sum of 7 and 4 = 11

Write a program to help Rohit achieve this for any given two numbers.

Note: Tile sign of the input numbers should be ignored.

i.e.

if the input numbers are 267 and 154, the sum of last two digits should be 11

if the input numbers are 267 and -154, the slim of last two digits should be 11

if the input numbers are -267 and 154, the sum of last two digits should be 11

if the input numbers are -267 and -154, the sum of last two digits should be 11

For example:

Input	Result
267 154	11
267 -154	11

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
a=int(input())
b=int(input())

d1=a%10
d2=abs(b)%10


print(d1+d2)
```

	Input	Expected	Got	
✔	267 154	11	11	✔
✔	267 -154	11	11	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 2 | Correct Mark 10.00 out of 10.00

Note:

Dont use if-else. Operators alone must be used .

A team from the Rotract club had planned to conduct a rally to create awareness among the Coimbatore people to donate blood. They conducted the rally successfully. Many of the Coimbatore people realized it and came forward to donate their blood to nearby blood banks. The eligibility criteria for donating blood are people should be above or equal to 18 and his/ her weight should be above 40. There was a huge crowd and staff in the blood bank found it difficult to manage the crowd. So they decided to keep a system and ask the people to enter their age and weight in the system. If a person is eligible he/she will be allowed inside.

Write a program and feed it to the system to find whether a person is eligible or not.

Input Format:

Input consists of two integers that correspond to the age and weight of a person respectively.

Output Format:

Display True(IF ELIGIBLE)

Display False (if not eligible)

Sample Input

19

45

Sample Output

True

For example:

Input	Result
18 40	False

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
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```
a=int(input())
b=int(input())
print(a>=18 and b>40)
```

	Input	Expected	Got	
✔	19 45	True	True	✔
✔	18 40	False	False	✔
✔	18 42	True	True	✔
✔	16 45	False	False	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 3 | Correct Mark 10.00 out of 10.00

You went on a tour to Ooty with your friends. As a part of the tour, you went boating with them. For the boat to remain stable, the number of people on one boat is restricted based on the weight of the people. You find that the boatman who is sailing your boat is so much greedy of money. For earning more, he takes too many people to travel in the boat at a time. So you want to check how many people can travel in the boat at a time so that the boat will not drown. Calculate the weight by considering the number of adults and number of children. Assume that an adult weighs 75 kg and children weigh 30 kg each. If the weight is normal, display Boat is stable, else display Boat will drown.

INPUT & OUTPUT FORMAT:

Input consists of 3 integers.

First input corresponds to the weight that the boat can handle.

Second input corresponds to the number of adults.

Third input corresponds to the number of children.

For example:

Input	Result
340 2 3	Boat is stable
600 7 4	Boat will drow

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
a=int(input())
b=int(input())
c=int(input())
d=b*75
e=c*30
f=d+e
print(a>f and "Boat is stable" or "Boat will drow")
```

	Input	Expected	Got	
✔	340 2 3	Boat is stable	Boat is stable	✔
✔	600 7 4	Boat will drow	Boat will drow	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 4 | Correct Mark 10.00 out of 10.00

In the 1800s, the battle of Troy was led by Hercules. He was a superstitious person. He believed that his crew can win the battle only if the total count of the weapons in hand is in multiple of 3 and the soldiers are in an even number of count. Given the total number of weapons and the soldier's count, Find whether the battle can be won or not according to Hercules's belief. If the battle can be won print True otherwise print False.

Input format:

Line 1 has the total number of weapons

Line 2 has the total number of Soldiers.

Output Format:

If the battle can be won print True otherwise print False.

Sample Input:

32

43

Sample Output:'

False

For example:

Input	Result
32 43	False

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
a=int(input())
b=int(input())

print( a%3==0 and b%2==0)
```

	Input	Expected	Got	
✔	32 43	False	False	✔

	Input	Expected	Got	
✔	273 7890	True	True	✔
✔	800 4590	False	False	✔
✔	6789 32996	True	True	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 5 | Correct Mark 10.00 out of 10.00

An online retailer sells two products: widgets and gizmos. Each widget weighs 75 grams. Each gizmo weighs 112 grams. Write a program that reads the number of widgets and the number of gizmos from the user. Then your program should compute and display the total weight of the parts.

Sample Input:

10

20

Sample Output:

The total weight of all these widgets and gizmos is 2990 grams.

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
a=int(input())
b=int(input())
c=a*75
d=b*112
e=c+d
print("The total weight of all these widgets and gizmos is",e,"grams.")
```

	Input	Expected	Got	
✔	10 20	The total weight of all these widgets and gizmos is 2990 grams.	The total weight of all these widgets and gizmos is 2990 grams.	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 6 | Correct Mark 10.00 out of 10.00

Complete the program to convert days into years, month and days. (Ignoring leap year and considering 1 month is 30 days)

Sample Test Cases

Test Case 1

Input

375

Output

YEARS: 1 MONTH: 0 DAYS: 10

Test Case 2

Input

200

Output

YEARS: 0 MONTH: 6 DAYS: 20

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
days=int(input())

years=days//365
days=days%365

months=days//30
days=days%30

print('YEARS:',years,'MONTH:',months,'DAYS:',days)
```

	Input	Expected	Got	
✔	375	YEARS: 1 MONTH: 0 DAYS: 10	YEARS: 1 MONTH: 0 DAYS: 10	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 7 | Correct Mark 10.00 out of 10.00

Write a simple python program to find the square root of a given floating point number. The output should be displayed with 3 decimal places.

Sample Input:

8.00

Sample Output:

2.828

For example:

Input	Result
8.00	2.828
14.00	3.742

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
a=float(input())
print("%.3f"%a**0.5)
```

	Input	Expected	Got	
✔	8.00	2.828	2.828	✔
✔	14.00	3.742	3.742	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 8 | Correct Mark 10.00 out of 10.00

Mr.Ram has been given a problem kindly help him to solve it. The input of the program is either 0 or 1. IF 0 is the input he should display "C" if 1 is the input it should display "D".There is a constraint that Mr. Ram should use either logical operators or arithmetic operators to solve the problem, not anything else.

Hint:

Use ASCII values of C and D.

Input Format:

An integer x, 0<=x<=1. .

Output Format:

output a single character "C" or "D"depending on the value of x.

```
Input 1:
0
Output 1:
C
```

```
Input 2:
1

Output 1:
D
```

For example:

Input	Result
0	C

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
n=int(input())
print(n==0 and'C'or'D')
```

	Input	Expected	Got	
✔	0	C	C	✔
✔	1	D	D	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 9 | Correct Mark 10.00 out of 10.00

Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of his basic salary, and his house rent allowance is 20% of his basic salary. Write a program to calculate his gross salary.

Sample Input:

10000

Sample Output:

16000

For example:

Input	Result
10000	16000

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
n=int(input())
a=(n+(0.4*n)+(0.2*n))
print(int(a))
```

	Input	Expected	Got	
✔	10000	16000	16000	✔
✔	20000	32000	32000	✔
✔	28000	44800	44800	✔
✔	5000	8000	8000	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 10 | Correct Mark 10.00 out of 10.00

Write a python program that takes a integer between 0 and 15 as input and displays the number of '1' s in its binary form.(Hint:use python bitwise operator.

Sample Input

3

Sample Output:

2

Explanation:

The binary representation of 3 is 011, hence there are 2 ones in it. so the output is 2.

For example:

Input	Result
3	2

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
num=int(input())
count=0
while num>0:
    count+=num&1
    num=num>>1
print(count)
```

	Input	Expected	Got	
✔	3	2	2	✔
✔	5	2	2	✔
✔	15	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Started on	Friday, 24 July 2026, 2:32 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 11:13 PM
Time taken	12 days 8 hours
Marks	2.00/10.00
Grade	**20.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

Write a program that returns the last digit of the given number. Last digit is being referred to the least significant digit i.e. the digit in the ones (units) place in the given number.

The last digit should be returned as a positive number.

For example,

if the given number is 197, the last digit is 7

if the given number is -197, the last digit is 7

For example:

Input	Result
197	7
-197	7

Answer: (penalty regime: 0 %)

```
a=int(input())

b=abs(a)%10
print(b)
```

	Input	Expected	Got	
✔	197	7	7	✔
✔	-197	7	7	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Write a program to find whether the given input number is Even.

If the given number is even, the function should return 2 else it should return 1.

Note: The number passed to the program can either be negative, positive or zero. Zero should be treated as Even.

For example:

Input	Result
100	2
1001	1

Answer: (penalty regime: 0 %)

```
n=int(input())
print(n%2==0 and "2" or "1")
```

	Input	Expected	Got	
✔	100	2	2	✔
✔	1001	1	1	✔
✔	0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Not answered Mark 0.00 out of 1.00

Justin is a carpenter who works on an hourly basis. He works in a company where he is paid Rs 50 for an hour on weekdays and Rs 80 for an hour on weekends. He works 10 hrs more on weekdays than weekends. If the salary paid for him is given, write a program to find the number of hours he has worked on weekdays and weekends.

Hint:

If the final result(hrs) are in -ve convert that to +ve using abs() function

The `abs()` function returns the absolute value of the given number.

```
number = -20
absolute_number = abs(number)
print(absolute_number)
# Output: 20
```

Sample Input:

450

Sample Output:

weekdays 10.38

weekend 0.38

For example:

Input	Result
450	weekdays 10.38 weekend 0.38

Answer: (penalty regime: 0 %)

```
1
```

Question 4 | Not answered Mark 0.00 out of 1.00

In department 54% are boys and 46% are girls and 8% are hostel (boys/girls). write a python code to print total no of boys, girls and hostel students in the specific format using modulo operator.

input: 1500

output: Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120

For example:

Input	Result
1500	Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120
1200	Total Students : 1200, Boys : 648, Girls : 552, Hostel : 96

Answer: (penalty regime: 0 %)

```
1
```

Question 5 | Not answered Mark 0.00 out of 1.00

Write a program to convert strings to an integer and float and display its type.

Sample Input:

10

10.9

Sample Output:

10,<class 'int'>

10.9,<class 'float'>

For example:

Input	Result
10 10.9	10,<class 'int'> 10.9,<class 'float'>

Answer: (penalty regime: 0 %)

```
1
```

Question 6 | Not answered Mark 0.00 out of 1.00

In a Lab 36% are Dell and 34% Lennovo and 28% are Acer and 2% are Samsung. write a python code to print total systems and brand wise count in the specific format using sep operator.

input: 150

output: Total System:150

Dell:54

Lennovo:51

Acer:42

Samsung:3

For example:

Input	Result
150	Total System:150 Dell:54 Lennovo:51 Acer:42 Samsung:3
250	Total System:250 Dell:90 Lennovo:85 Acer:70 Samsung:5

Answer: (penalty regime: 0 %)

```
1
```

Question 7 | Not answered Mark 0.00 out of 1.00

In many jurisdictions, a small deposit is added to drink containers to encourage people to recycle them. In one particular jurisdiction, drink containers holding one liter or less have a $0.10 deposit and drink containers holding more than one liter have a $0.25 deposit. Write a program that reads the number of containers of each size(less and more) from the user. Your program should continue by computing and displaying the refund that will be received for returning those containers. Format the output so that it includes a dollar sign and always displays exactly two decimal places.

Sample Input

10

20

Sample Output

Your total refund will be $6.00.

For example:

Input	Result
20 20	Your total refund will be $7.00.

Answer: (penalty regime: 0 %)

```
1
```

Question 8 | Not answered Mark 0.00 out of 1.00

Write a program that returns the second last digit of the given number. Second last digit is being referred 10the digit in the tens place in the given number.

For example, if the given number is 197, the second last digit is 9.

Note1 - The second last digit should be returned as a positive number. i.e. if the given number is -197, the second last digit is 9.

Note2 - If the given number is a single digit number, then the second last digit does not exist. In such cases, the program should return -1. i.e. if the given number is 5, the second last digit should be returned as -1

For example:

Input	Result
197	9
-197	9
5	-1

Answer: (penalty regime: 0 %)

```
1
```

Question 9 | Not answered Mark 0.00 out of 1.00

In a Logistic the Parcels to be delivered in 4 locations (1st locaion 20%, 2nd location 40%, 3rd location 30% and 4th location 10%). write a python code to find the total no. of parcels after the delivery in 2 locations . use a format() to print the no of parcels delivered in in each location

Input:

250

output:

Total Parcels is 250

1st Location 50 parcels

2nd Location 100 parcels

3rd Location 75 parcels

4th Location 25 parcels

For example:

Input	Result
250	Total Parcels is 250 1st Location 50 parcels 2nd Location 100 parcels 3rd Location 75 parcels 4th Location 25 parcels
400	Total Parcels is 400 1st Location 80 parcels 2nd Location 160 parcels 3rd Location 120 parcels 4th Location 40 parcels

Answer: (penalty regime: 0 %)

```
1
```

Question 10 | Not answered Mark 0.00 out of 1.00

Alfred buys an old scooter for Rs. X and spends Rs. Y on its repairs. If he sells the scooter for Rs. Z (Z>X+Y). Write a program to help Alfred to find his gain percent. Get all the above-mentioned values through the keyboard and find the gain percent.

Input Format:

The first line contains the Rs X

The second line contains Rs Y

The third line contains Rs Z

Sample Input:

10000

250

15000

Sample Output:

46.34 is the gain percent.

For example:

Input	Result
45500 500 60000	30.43 is the gain percent.

Answer: (penalty regime: 0 %)

```
1
```

Started on	Thursday, 13 August 2026, 3:30 PM
State	Finished
Completed on	Monday, 17 August 2026, 11:59 PM
Time taken	4 days 8 hours
Marks	1.00/10.00
Grade	**10.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

A triangle can be classified based on the lengths of its sides as equilateral, isosceles or scalene. All three sides of an equilateral triangle have the same length. An isosceles triangle has two sides that are the same length, and a third side that is a different length. If all of the sides have different lengths then the triangle is scalene.

Write a program that reads the lengths of the three sides of a triangle from the user. Then display a message that states the triangle's type.

Sample Input 1

60

60

60

Sample Output 1

That's a equilateral triangle

Sample Input 2

40

40

80

Sample Output 2

That's a isosceles triangle

Sample Input 3

50

60

70

Sample Output 3

That's a scalene triangle

For example:

Input	Result
60 60 60	That's a equilateral triangle
40 40 80	That's a isosceles triangle

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
if a==b and b==c:
  print("That's a equilateral triangle")
elif a==b or b==c or a==c:
  print("That's a isosceles triangle")
else:
  print("That's a scalene triangle")
```

	Input	Expected	Got	
✔	60 60 60	That's a equilateral triangle	That's a equilateral triangle	✔
✔	40 40 80	That's a isosceles triangle	That's a isosceles triangle	✔
✔	50 60 70	That's a scalene triangle	That's a scalene triangle	✔
✔	50 50 80	That's a isosceles triangle	That's a isosceles triangle	✔
✔	10 10 10	That's a equilateral triangle	That's a equilateral triangle	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Incorrect Mark 0.00 out of 1.00

Write a Python program that accepts three parameters. The first parameter is an integer. The second is one of the following mathematical operators: +, -, /, or *. The third parameter will also be an integer.

The function should perform a calculation and return the results. For example, if the function is passed 6 and 4, it should return 24.

Sample Input Format:

11

+

14

Sample Output Format:

25

Answer: (penalty regime: 0 %)

```
a=int(input())
b=input()
c=int(input())

if b=="+":
    print(a+c)
elif b=="-":
    print(a-c)
elif
```

Syntax Error(s)

```
  File "__tester__.python3", line 9
    elif
        ^
SyntaxError: invalid syntax
```

Incorrect

Marks for this submission: 0.00/1.00.

Question 3 | Not answered Mark 0.00 out of 1.00

Write a program that accepts 5 inputs and returns the count of how many of those 5 are odd.

For example,

If the five inputs are 12, 17, 19, 14, and 115, there are three odd numbers 17, 19 and 115. So, the program must return 3.

Similarly,

If the five inputs are 15, 0, -12, 19, and 28, there are two odd numbers 15 and 19. So, the program must return 2.

Observe that zero is considered an even number.

For example:

Input	Result
12 17 19 14 115	3
15 0 -12 19 28	2

Answer: (penalty regime: 0 %)

```
1
```

Question 4 | Not answered Mark 0.00 out of 1.00

Write a program to calculate and print the Electricity bill where the unit consumed by the user is given from test case. It prints the total amount the customer has to pay. The charge are as follows:

Unit	Charge / Unit
Upto 199	@1.20
200 and above but less than 400	@1.50
400 and above but less than 600	@1.80
600 and above	@2.00

If bill exceeds Rs.400 then a surcharge of 15% will be charged and the minimum bill should be of Rs.100/-

Sample Test Cases

Test Case 1

Input

50

Output

100.00

Test Case 2

Input

300

Output

517.50

For example:

Input	Result
100.00	120.00
500	1035.00

Answer: (penalty regime: 0 %)

```
1
```

Question 5 | Not answered Mark 0.00 out of 1.00

Most years have 365 days. However, the time required for the Earth to orbit the Sun is actually slightly more than that. As a result, an extra day, February 29, is included in some years to correct for this difference. Such years are referred to as leap years. The rules for determining whether or not a year is a leap year follow:

• Any year that is divisible by 400 is a leap year.

• Of the remaining years, any year that is divisible by 100 is not a leap year.

• Of the remaining years, any year that is divisible by 4 is a leap year.

• All other years are not leap years.

Write a program that reads a year from the user and displays a message indicating whether or not it is a leap year.

Sample Input 1

1900

Sample Output 1

1900 is not a leap year.

Sample Input 2

2000

Sample Output 2

2000 is a leap year.

Answer: (penalty regime: 0 %)

```
1
```

Question 6 | Not answered Mark 0.00 out of 1.00

The Chinese zodiac assigns animals to years in a 12 year cycle. One 12 year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the dragon, and 1999 being another year of the hare.

Year Animal

2000 Dragon

2001 Snake

2002 Horse

2003 Sheep

2004 Monkey

2005 Rooster

2006 Dog

2007 Pig

2008 Rat

2009 Ox

2010 Tiger

2011 Hare

Write a program that reads a year from the user and displays the animal associated with that year. Your program should work correctly for any year greater than or equal to zero, not just the ones listed in the table.

Sample Input 1

2010

Sample Output 1

2010 is the year of the Tiger.

Sample Input 2

2020

Sample Output 2

2020 is the year of the Rat.

Answer: (penalty regime: 0 %)

```
1
```

Question 7 | Not answered Mark 0.00 out of 1.00

IN / OUT

Ms. Sita, the faculty handling programming lab for you is very strict. Your seniors have told you that she will not allow you to enter the week's lab if you have not completed atleast half the number of problems given last week. Many of you didn't understand this statement and so they requested the good programmers from your batch to write a program to find whether a student will be allowed into a week's lab given the number of problems given last week and the number of problems solved by the student in that week.

Input Format:

Input consists of 2 integers.

The first integer corresponds to the number of problems given and the second integer corresponds to the number of problems solved.

Output Format:

Output consists of the string “IN” or “OUT”.

Sample Input and Output:

Input

8

3

Output

OUT

For example:

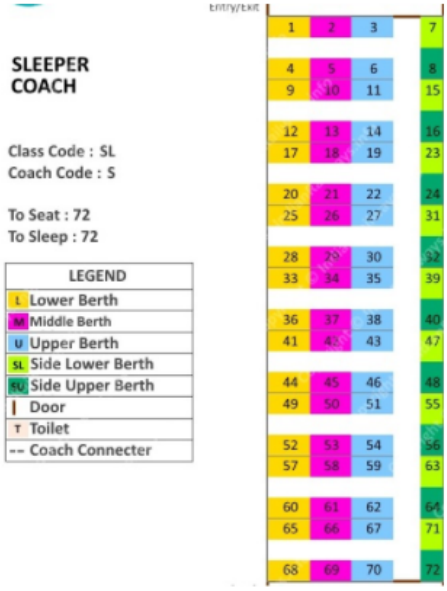
Input	Result
8 3	OUT

Answer: (penalty regime: 0 %)

```
1
```

Question 8 | Not answered Mark 0.00 out of 1.00

Write a program to determine the type of berth when the seat / berth number in the train is given.



Input Format:

Input consists of a single integer. Assume that the range of input is between 1 and 72.

Output Format:

Output consists of a single string. [Upper or Middle or Lower or Side Lower or Side Upper]

Sample Input 1:

9

Sample Output 1:

Lower Berth

Answer: (penalty regime: 0 %)

```
1
```

Question 9 | Not answered Mark 0.00 out of 1.00

A certain type of steel is used to test and give grade according to the following conditions.

1. Hardness of the steel must be greater than 50
2. Carbon content of the steel must be less than 0.7
3. Tensile strength must be greater than 5600

The grades awarded are as follows:

- Grade is 10 if all three conditions are met
- Grade is 9 if conditions (1) and (2) are met
- Grade is 8 if conditions (2) and (3) are met
- Grade is 7 if conditions (1) and (3) are met
- Grade is 6 if only one condition is met
- Grade is 5 if none of the three conditions are met

Write a program to display the grade of the steel, based on the values of hardness, carbon content and tensile strength of the steel, given by the user.

Input

53

0.6

5602

Output:

10

Answer: (penalty regime: 0 %)

```
1
```

Question 10 | Not answered Mark 0.00 out of 1.00

Given an integer N, check whether N the given number can be made a perfect square after adding 1 to it.

Input Format:

Single integer input.

Output Format:

Yes or No.

Example Input:

24

Output:

Yes

Example Input:

26

Output:

No

For example:

Input	Result
24	Yes

Answer: (penalty regime: 0 %)

```
1
```

Started on	Thursday, 20 August 2026, 4:11 PM
State	Finished
Completed on	Thursday, 20 August 2026, 4:36 PM
Time taken	24 mins 36 secs
Marks	5.00/5.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

Example 1:

Input:

69

Output:

true

Example 2:

Input:

88

Output:

true

Example 3:

Input:

962

Output:

false

Example 4:

Input:

1

Output:

true

For example:

Test	Result
`print(Strobogrammatic(69))`	`True`
`print(Strobogrammatic(962))`	`False`

Answer: (penalty regime: 0 %)

Reset answer

```
def Strobogrammatic(n):
    s=str(n)
    d={'0':'0','1':'1','8':'8','6':'9','9':'6'}
    for i in range(len(s)):
        if s[i] not in d:
            return False
        if d[s[i]]!=s[len(s)-1-i]:
            return False
    return True
    n=int(input())
    print(Strobogrammatic(n))
```

	Test	Expected	Got	
✔	`print(Strobogrammatic(69))`	`True`	`True`	✔
✔	`print(Strobogrammatic(88))`	`True`	`True`	✔
✔	`print(Strobogrammatic(962))`	`False`	`False`	✔
✔	`print(Strobogrammatic(66))`	`False`	`False`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all

the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

For example:

Test	Result
`print(christmasDiscount(578))`	12

Answer: (penalty regime: 0 %)

Reset answer

```
def christmasDiscount(n):
    return sum(int(x) for x in str(n) if x in "2357")

```

	Test	Expected	Got	
✔	`print(christmasDiscount(578))`	12	12	✔
✔	`print(christmasDiscount(57))`	12	12	✔
✔	`print(christmasDiscount(222))`	6	6	✔
✔	`print(christmasDiscount(77777))`	35	35	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

For example:

Test	Result
`print(checkUgly(6))`	`ugly`
`print(checkUgly(21))`	`not ugly`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
    while(n%2)==0:
        n=n/2
    while(n%3)==0:
        n=n/3
    while(n%5)==0:
        n=n/5
    if(n==1):
        return("ugly")
    else:
        return("not ugly")

```

	Test	Expected	Got	
✔	`print(checkUgly(6))`	`ugly`	`ugly`	✔
✔	`print(checkUgly(21))`	`not ugly`	`not ugly`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

Answer: (penalty regime: 0 %)

Reset answer

```
def coinChange(n):
    c=0
    sum=0
    while(sum<n):
        if(sum+4<=n):
            sum+=4
            c+=1
        elif(sum+3<=n):
            sum+=3
            c+=1
        elif(sum+2<=n):
            sum+=2
            c+=1
        elif(sum+1<=n):
            sum+=1
            c+=1
    return(c)

```

	Test	Expected	Got	
✔	print(coinChange(16))	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output format:

11106

For example:

Test	Result
`print(Summation(8))`	9872

Answer: (penalty regime: 0 %)

Reset answer

```
def Summation(n):
    x=str(n)
    s=int(x)+int(x*2)+int(x*3)+int(x*4)
    return(s)
```

	Test	Expected	Got	
✔	`print(Summation(8))`	9872	9872	✔
✔	`print(Summation(10))`	10203040	10203040	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Thursday, 27 August 2026, 3:20 PM
State	Finished
Completed on	Thursday, 27 August 2026, 4:01 PM
Time taken	41 mins 31 secs
Marks	5.00/5.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

Complete a Recursive Function to find if a given number N can be expressed as a sum of two prime numbers.

Note: YOU MUST OPTIMIZE the logic to find whether a number is prime or not, as very large prime numbers are provided as input. If the logic is not optimized your program will NOT get executed within the given time limit.

Input Format:

First line contains number N.

Output Format:

Return either yes or no.

Boundary Conditions / Constraints:

3 <= N <= 10^9

Example Input/Output 1:

Input:

20

Output:

yes

Input:

23

Ouput:

no

Explanation:

20 can be expressed as 17+3

23 cannot be expressed as sum of two primes

For example:

Test	Result
print(checkPrimeSum(20))	yes
print(checkPrimeSum(23))	no

Answer: (penalty regime: 0 %)

Reset answer

```
def checkPrime(n):
    if n<2:
        return False
    for i in range(2,int(n**0.5)+1):
        if n%i==0:
            return False
    return True

def checkPrimeSum(n,i=2):
    if (i>n//2):
        return "no"
    if checkPrime(i) and checkPrime(n-i):
        return "yes"
    return checkPrimeSum (n,i+1)

```

	Test	Expected	Got	
✔	`print(checkPrimeSum(20))`	`yes`	`yes`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Complete the recursive function to return Binary Equivalent of an Integer using Recursion.

Sample Test Cases

Test Case 1

Input

10

Output

1010

Test Case 2

Input

257

Output

100000001

For example:

Test	Result
print(binayNumber(10))	1010
print(binayNumber(257))	100000001

Answer: (penalty regime: 0 %)

Reset answer

```
def binayNumber(n):
    if n<2:
        return n
    return binayNumber(n//2)*10+n%2

```

	Test	Expected	Got	
✔	print(binayNumber(10))	1010	1010	✔
✔	print(binayNumber(257))	100000001	100000001	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

The notion of a palindrome was introduced previously. In this exercise you will write a recursive function that determines whether or not a string is a palindrome. The empty string is a palindrome, as is any string containing only one character. Any longer string is a palindrome if its first and last characters match, and if the string formed by removing the first and last characters is also a palindrome.

Write a program that reads a string from the user and uses your recursive function to determine whether or not it is a palindrome. Then your program should display an appropriate message for the user.

Sample Input

malayalam

Sample Output

That was a palindrome!

Sample Input

madan

Sample Output

That is not a palindrome.

Answer: (penalty regime: 0 %)

Reset answer

```
def isPalindrome(s):
    if len(s)<=1:
        return 1
    x=s[::-1]

    if(x==s):
        return 1
    else:
        return 0
line=input()
if isPalindrome(line):
    print("That was a palindrome!")
else:
    print("That is not a palindrome.")
```

	Input	Expected	Got	
✔	malayalam	That was a palindrome!	That was a palindrome!	✔
✔	madan	That is not a palindrome.	That is not a palindrome.	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

Euclid was a Greek mathematician who lived approximately 2,300 years ago. His algorithm for computing the greatest common divisor of two positive integers, a and b, is both efficient and recursive. It is outlined below:

If b is 0 then
  return a
Else
  Set c equal to the remainder when a is divided by b
  Return the greatest common divisor of b and c

Write a Recursive funtion that implements Euclid's algorithm and uses it to determine the greatest common divisor of two integers entered by the user. Test your program with some very large integers. The result will be computed quickly, even for huge numbers consisting of hundreds of digits, because Euclid's algorithm is extremely efficient.

For example:

Test	Result
`print(gcd(8, 12))`	`4`
`print(gcd(720, 1000))`	`40`

Answer: (penalty regime: 0 %)

Reset answer

```
def gcd(a,b):
    if b==0:
        return a
    return gcd(b,a%b)
```

	Test	Expected	Got	
✔	`print(gcd(8, 12))`	`4`	`4`	✔
✔	`print(gcd(720, 1000))`	`40`	`40`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Given an integer number and you have to count the digits using recursion using Python program. In this program, you will be reading an integer number and counting the total digits, using a function countDigits() which will take a number as an argument and return the count after recursion process.

Input Format: The first and only line of the input contains a single integer n

Output Format: Output a single line denoting the number of digits in n.

For example:

Test	Result
`print(countDigits(800))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def countDigits(n):
    count=0
    while(n>0):
        n=n//10
        count+=1
        countDigits(n)
    return count

```

	Test	Expected	Got	
✔	`print(countDigits(12345))`	5	5	✔
✔	`print(countDigits(800))`	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Thursday, 1 October 2026, 4:07 PM
State	Finished
Completed on	Monday, 5 October 2026, 12:51 PM
Time taken	3 days 20 hours
Marks	10.00/10.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1

2

2

3

4

Output:

1 2 3 4

Example Input:

6

1

1

2

2

3

3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4
6 1 1 2 2 3 3	1 2 3

Answer: (penalty regime: 0 %)

```
1 import sys
```

```
2
3  data=sys.stdin.read().split()
4  n=int(data[0])
5  a=list(map(int,data[1:n+1]))
6
7  seen=set()
8  res=[]
9  for x in a:
10     if x not in seen:
11         seen.add(x)
12         res.append(x)
13 print(*res)
```

	Input	Expected	Got	
✔	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✔
✔	6 1 1 2 2 3 3	1 2 3	1 2 3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Complete the program to count frequency of each element of an array. Frequency of a particular element will be printed once.

Sample Test Cases

Test Case 1

Input

7
23
45
23
56
45
23
40

Output

23 occurs 3 times
45 occurs 2 times
56 occurs 1 times
40 occurs 1 times

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n=int(data[0])
a=data[1:n+1]

freq={}
for x in a:
    freq[x]=freq.get(x,0) +1

for x,c in freq.items():
    print(f"{x} occurs {c} times")

```

	Input	Expected	Got	
✔	7 23 45 23 56 45 23 40	23 occurs 3 times 45 occurs 2 times 56 occurs 1 times 40 occurs 1 times	23 occurs 3 times 45 occurs 2 times 56 occurs 1 times 40 occurs 1 times	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

An array is monotonic if it is either **monotone increasing** or **monotone decreasing**.
An array A is monotone increasing if for all i <= j, **A[i] <= A[j]**. An array A is monotone decreasing if for all i <= j, **A[i] >= A[j]**.

Write a program if n array is monotonic or not. Print "True" if is monotonic or "False" if it is not. Array can be monotone increasing or decreasing.

Input Format:

First line n-get number of elements

Next n Lines is the array of elements

Output Format:

True ,if array is monotone increasing or decreasing.

otherwise False is printed

Sample Input1

4

5

6

7

8

Sample Output1

True

Sample Input2

4

6

5

4

3

Sample Output2

True

Sample Input 3

4

6

7

8

7

Sample Output3

False

For example:

Input	Result
4 6 5 4 3	True

Answer: (penalty regime: 0 %)

```
n=int(input())
```

```
a=[int(input()) for _ in range(n)]
inc=all(a[i] <= a[i+1] for i in range(n-1))
dec=all(a[i] >= a[i+1] for i in range(n-1))
print("True" if inc or dec else "False")

```

	Input	Expected	Got	
✔	4 6 5 4 3	True	True	✔
✔	4 3 5 7 4	False	False	✔
✔	4 1 6 9 2	False	False	✔
✔	4 9 6 4 2	True	True	✔
✔	3 2 1 4	False	False	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

The program must accept **N** integers and an integer **K** as the input. The program must print every K integers in descending order as the output.

Note: If **N % K != 0**, then sort the final N%K integers in descending order.

Boundary Condition(s):

1 <= N <= 10^4
-99999 <= Array Element Value <= 99999

Input Format:

The first line contains the values of N and K separated by a space.
The second line contains N integers separated by space(s).

Output Format:

The first line contains N integers.

Example Input/Output 1:

Input:

7 3
48 541 23 68 13 41 6

Output:

541 48 23 68 41 13 6

Explanation:

The first three integers are 48 541 23, after sorting in descending order the integers are **541 48 23**.
The second three integers are 68 13 41, after sorting in descending order the integers are **68 41 13**.
The last integer is **6**.
The integers are **541 48 23 68 41 13 6**
Hence the output is **541 48 23 68 41 13 6**.

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n,k=int(data[0]),int(data[1])
a=list(map(int, data[2:2+n]))

res=[]
for  i in range(0,n,k):
    res.extend(sorted(a[i:i+k], reverse=True))

print(*res)
```

	Input	Expected	Got	
✔	7 3 48 541 23 68 13 41 6	541 48 23 68 41 13 6	541 48 23 68 41 13 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Given an integer n, return an list of length n + 1 such that for each i (0 <= i <= n), ans[i] is the number of 1's in the binary representation of i.

Example:

```
Input: n = 2
Output: [0,1,1]
Explanation:
0 --> 0
1 --> 1
2 --> 10
```

Example2:

```
Input: n = 5
Output: [0,1,1,2,1,2]
Explanation:
0 --> 0
1 --> 1
2 --> 10
3 --> 11
4 --> 100
5 --> 101
```

Note: Complete the given function alone

For example:

Test	Result
print(CountingBits(5))	[0, 1, 1, 2, 1, 2]

Answer: (penalty regime: 0 %)

Reset answer

```
def CountingBits(n):
    ans = [0] * (n+1)
    for i in range(1,n+1):
        ans[i]=ans[i>>1] + (i&1)
    return ans
```

	Test	Expected	Got	
✔	print(CountingBits(2))	[0, 1, 1]	[0, 1, 1]	✔
✔	print(CountingBits(5))	[0, 1, 1, 2, 1, 2]	[0, 1, 1, 2, 1, 2]	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 6 | Correct Mark 1.00 out of 1.00

Given two arrays of positive integers, for each element in the second array, find the total number of elements in the first array which are *less than or equal to* that element. Store the values determined in an array.

For example, if the first array is *[1, 2, 3]* and the second array is *[2, 4]*, then there are *2* elements in the first array *less than or equal to 2.* There are *3* elements in the first array which are *less than or equal to 4*. We can store these answers in an array, *answer = [2, 3]*.

Program Description

The program must return an array of *m* positive integers, one *for each maxes[i]* representing the total number of elements *nums[j]* satisfying *nums[j] ≤ maxes[i]* where *0 ≤ j < n* and *0 ≤ i < m,* in the given order.

The program has the following:

nums[nums[0],...nums[n-1]]: first array of positive integers

maxes[maxes[0],...maxes[n-1]]: second array of positive integers

Constraints

- $2 \le n, m \le 10^5$
- $1 \le nums[j] \le 10^9$, where $0 \le j < n$.
- $1 \le maxes[i] \le 10^9$, where $0 \le i < m$.

Input Format For Custom Testing

Input from stdin will be processed as follows and passed to the program.

The first line contains an integer *n*, the number of elements in *nums*.
The next *n* lines each contain an integer describing *nums[j]* where *0 ≤ j < n.*
The next line contains an integer *m*, the number of elements in *maxes*.
The next *m* lines each contain an integer describing *maxes[i]* where *0 ≤ i < m.*

Sample Case 0

Sample Input 0

```
4
1
4
2
4
2
3
5
```

Sample Output 0

```
2
4
```

Explanation 0

We are given *n = 4*, *nums = [1, 4, 2, 4]*, *m = 2*, and *maxes = [3, 5]*.

1. For *maxes[0] = 3*, we have *2* elements in *nums* (*nums[0] = 1* and *nums[2] = 2*) that are ≤ *maxes[0]*.
2. For *maxes[1] = 5*, we have *4* elements in *nums* (*nums[0] = 1*, *nums[1] = 4*, *nums[2] = 2*, and *nums[3] = 4*) that are ≤ *maxes[1]*.

Thus, the program returns the array *[2, 4]* as the answer.

Sample Case 1

Sample Input 1

```
5
2
10
5
4
8
4
3
1
7
8
```

Sample Output 1

```
1
0
3
4
```

Explanation 1

We are given, *n = 5*, *nums = [2, 10, 5, 4, 8]*, *m = 4*, and *maxes = [3, 1, 7, 8]*.

1. For *maxes[0] = 3*, we have *1* element in *nums* (*nums[0] = 2*) that is ≤ *maxes[0]*.
2. For *maxes[1] = 1*, there are *0* elements in *nums* that are ≤ *maxes[1]*.
3. For *maxes[2] = 7*, we have *3* elements in *nums* (*nums[0] = 2*, *nums[2] = 5*, and *nums[3] = 4*) that are ≤ *maxes[2]*.
4. For *maxes[3] = 8*, we have *4* elements in *nums* (*nums[0] = 2*, *nums[2] = 5*, *nums[3] = 4*, and *nums[4] = 8*) that are ≤ *maxes[3]*.

Thus, the program returns the array *[1, 0, 3, 4]* as the answer.

Answer: (penalty regime: 0 %)

```
import sys
from bisect import bisect_right

data= sys.stdin.read().split()
n=int(data[0])
nums=sorted(map(int,data[1:1+n]))
m=int(data[1+n])
maxes=list(map(int,data[2+n:2+n+m]))

for x in maxes:
    print(bisect_right(nums,x))
```

	Input	Expected	Got	
✔	4 1 4 2 4 2 3 5	2 4	2 4	✔
✔	5 2 10 5 4 8 4 3 1 7 8	1 0 3 4	1 0 3 4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 7 | Correct Mark 1.00 out of 1.00

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the pth element of the list, sorted ascending. If there is no p^{th} element, return 0.

Example

n = 20

p = 3

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if p = 3, then 4 is returned. If p > 6, 0 would be returned.

Constraints

$1 \le n \le 10^{15}$

$1 \le p \le 10^{9}$

The first line contains an integer n, the number to factor.

The second line contains an integer p, the 1-based index of the factor to return.

Sample Case 0

Sample Input 0

10

3

Sample Output 0

5

Explanation 0

Factoring n = 10 results in {1, 2, 5, 10}. Return the p = 3rd factor, 5, as the answer.

Sample Case 1

Sample Input 1

10

5

Sample Output 1

0

Explanation 1

Factoring n = 10 results in {1, 2, 5, 10}. There are only 4 factors and p = 5, therefore 0 is returned as the answer.

Sample Case 2

Sample Input 2

1

1

Sample Output 2

1

Explanation 2

Factoring n = 1 results in {1}. The p = 1st factor of 1 is returned as the answer.

For example:

Input	Result
10 3	5
10 5	0

Input	Result
1 1	1

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n=int(data[0])
p=int(data[1])

small=[]
large=[]
i=1
while i*i <= n:
    if n%i == 0:
        small.append(i)
        if i != n // i:
            large.append(n//i)
    i+=1
factors=small+large[::-1]
print(factors[p-1] if p <=len(factors) else 0)
```

	Input	Expected	Got	
✔	10 3	5	5	✔
✔	10 5	0	0	✔
✔	1 1	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 8 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[i] - A[j] = k, i != j.

Input Format

1. First line is number of test cases T. Following T lines contain:
2. N, followed by N integers of the array
3. The non-negative integer k

Output format

Print 1 if such a pair exists and 0 if it doesn't.

Example

Input

1

3

1

3

5

4

Output:

1

Input

1

3

1

3

5

99

Output

0

For example:

Input	Result
1 3 1 3 5 4	1
1 3 1 3 5 99	0

Answer: (penalty regime: 0 %)

```
1 import sys
2
```

```
data=sys.stdin.read().split()
pos=0
t=int(data[pos]); pos+=1

for _ in range(t):
    n=int(data[pos]); pos +=1
    a=list(map(int,data[pos:pos+n])); pos +=n
    k=int(data[pos]); pos+=1

    seen=set()
    found=0
    for x in a:
        if(x-k) in seen or (x+k) in seen:
            found = 1
            break
        seen.add(x)

    print(found)
```

	Input	Expected	Got	
✔	1 3 1 3 5 4	1	1	✔
✔	1 3 1 3 5 99	0	0	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 9 | Correct Mark 1.00 out of 1.00

Given a matrix mat where every row is sorted in **strictly increasing** order, return the **smallest common element** in all rows.

If there is no common element, return -1.

Example 1:

Input:

4 5

1 2 3 4 5

2 4 5 8 10

3 5 7 9 11

1 3 5 7 9

Output:

5

Constraints:

- 1 <= mat.length, mat[i].length <= 500
- 1 <= mat[i][j] <= 10^4
- mat[i] is sorted in strictly increasing order.

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
r,c=int(data[0]),int(data[1])
vals=list(map(int,data[2:2+r*c]))

count = [0] * 10001
for i in range(r):
    for x in vals[i*c:(i+1)*c]:
        count[x]+=1

ans = -1
for v in range(1,10001):
    if count[v] == r:
        ans =  v
        break
print(ans)
```

	Input	Expected	Got	
✔	4 5 1 2 3 4 5 2 4 5 8 10 3 5 7 9 11 1 3 5 7 9	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 10 | Correct Mark 1.00 out of 1.00

Assume you have an array of length ***n*** initialized with all **0**'s and are given ***k*** update operations.

Each operation is represented as a triplet: **[startIndex, endIndex, inc]** which increments each element of subarray **A[startIndex ... endIndex]** (startIndex and endIndex inclusive) with **inc**.

Return the modified array after all ***k*** operations were executed.

Example:

Input:

5

3

1 3 2

2 4 3

0 2 -2

Output:

-2 0 3 5 3

Explanation:

Initial state:

length = 5, updates = [[1,3,2],[2,4,3],[0,2,-2]]

[0,0,0,0,0]

After applying operation [1,3,2]:

[0,2,2,2,0]

After applying operation [2,4,3]:

[0,2,5,5,3]

After applying operation [0,2,-2]:

[-2,0,3,5,3]

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n=int(data[0])
k=int(data[1])

diff=[0] * (n+1)
pos=2
for _ in range(k):
    s=int(data[pos])
    e=int(data[pos+1])
    inc=int(data[pos+2])
    pos +=3
    diff[s] +=inc
    diff[e+1]-= inc

res=[]
cur=0
for i in range(n):
```

```
    cur += diff[i]
    res.append(cur)

print(*res)
```

	Input	Expected	Got	
✔	5 3 1 3 2 2 4 3 0 2 -2	-2 0 3 5 3	-2 0 3 5 3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 7 October 2026, 10:44 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 11:38 PM
Time taken	54 mins 17 secs
Marks	10.00/10.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

Check if a set is a subset of another set.

Example:

Sample Input1:

mango apple

mango orange

mango

output1:

yes

set3 is subset of set1 and set2

```
input2:
```

```
mango orange
```

```
banana orange
```

```
grapes
```

```
output2:
```

no

For example:

Test	Input	Result
1	mango apple mango orange mango	yes set3 is subset of set1 and set2
2	mango orange banana orange grapes	No

Answer: (penalty regime: 0 %)

```
set1=set(input().split())
set2=set(input().split())
set3=set(input().split())

if set3.issubset(set1) and set3.issubset(set2):
 print('yes')
 print('set3 is subset of set1 and set2')
else:
 print('No')

```

	Test	Input	Expected	Got	
✔	1	mango apple mango orange mango	yes set3 is subset of set1 and set2	yes set3 is subset of set1 and set2	✔
✔	2	mango orange banana orange grapes	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1 2 2 3 4

Output:

1 2 3 4

Example Input:

6

1 1 2 2 3 3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()


n=int(data[0])
arr=data[1:n+1]

result=list(dict.fromkeys(arr))
print(*result)
```

	Input	Expected	Got	
✔	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✔
✔	6 1 1 2 2 3 3	1 2 3	1 2 3	✔
✔	5 11 22 11 22 11	11 22	11 22	✔
✔	10 1 2 3 4 5 1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	✔

Passed all tests! ✔

Correct

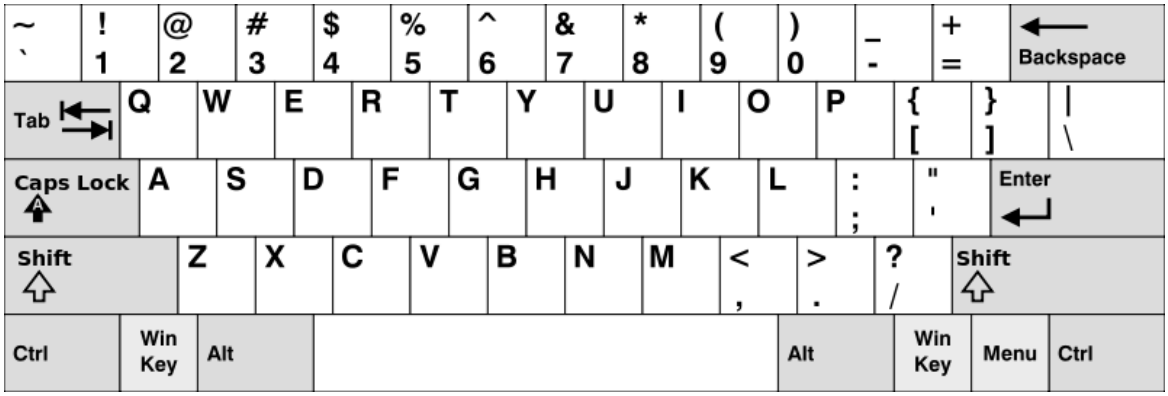
Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n=int(data[0])
words=data[1:n+1]

rows=[set('qwertyuiop'),set('asdfghjkl'),set('zxcvbnm')]

found=False

```

```
for w in words:
    letters=set(w.lower())
    if  any(letters <=row for row in rows):
        print(w)
        found=True

if not found:
        print('No words')
```

	Input	Expected	Got	
✔	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✔
✔	1 omk	No words	No words	✔
✔	2 adsfd afd	adsfd afd	adsfd afd	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

You are given an integer tuple `nums` containing distinct numbers. Your task is to perform a sequence of operations on this tuple until it becomes empty. The operations are defined as follows:

1. If the first element of the tuple has the smallest value in the entire tuple, remove it.
2. Otherwise, move the first element to the end of the tuple.

You need to return an integer denoting the number of operations required to make the tuple empty.

Constraints

- The input tuple `nums` contains distinct integers.
- The operations must be performed using tuples and sets to maintain immutability and efficiency.
- Your function should accept the tuple `nums` as input and return the total number of operations as an integer.

Example:

Input: nums = (3, 4, -1)
Output: 5

Explanation:
Operation 1: [3, 4, -1] -> First element is not the smallest, move to the end -> [4, -1, 3]
Operation 2: [4, -1, 3] -> First element is not the smallest, move to the end -> [-1, 3, 4]
Operation 3: [-1, 3, 4] -> First element is the smallest, remove it -> [3, 4]
Operation 4: [3, 4] -> First element is the smallest, remove it -> [4]
Operation 5: [4] -> First element is the smallest, remove it -> []
Total operations: 5

For example:

Test	Result
`print(count_operations((3, 4, -1)))`	5

Answer: (penalty regime: 0 %)

Reset answer

```
from collections import deque

def count_operations(nums):
    order=sorted(nums)
    d=deque(nums)
    i=0
    ops=0
    while d:
        if d[0] == order[i]:
            d.popleft()
            i+=1
        else:
            d.append(d.popleft())
        ops += 1
    return ops
```

	Test	Expected	Got	
✔	`print(count_operations((3, 4, -1)))`	5	5	✔

	Test	Expected	Got	
✔	`print(count_operations((1, 2, 3, 4, 5)))`	5	5	✔
✔	`print(count_operations((5, 4, 3, 2, 1)))`	15	15	✔
✔	`print(count_operations((42, )))`	1	1	✔
✔	`print(count_operations((-2, 3, -5, 4, 1)))`	11	11	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Coders here is a simple task for you, Given string str. Your task is to check whether it is a binary string or not by using python set.

Examples:

Input: str = "01010101010"

Output: Yes

Input: str = "REC101"

Output: No

For example:

Input	Result
01010101010	Yes
010101 10101	No

Answer: (penalty regime: 0 %)

```
s=input()

if set(s) <= {'0','1'}:
    print('Yes')
else:
    print('No')
```

	Input	Expected	Got	
✔	01010101010	Yes	Yes	✔
✔	REC123	No	No	✔
✔	010101 10101	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 6 | Correct Mark 1.00 out of 1.00

The **DNA sequence** is composed of a series of nucleotides abbreviated as `'A'`, `'C'`, `'G'`, and `'T'`.

- For example, `"ACGAATTCCG"` is a **DNA sequence**.

When studying **DNA**, it is useful to identify repeated sequences within the DNA.

Given a string `s` that represents a **DNA sequence**, return all the **`10`-letter-long** sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in **any order**.

Example 1:

```
Input: s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"
Output: ["AAAAACCCCC","CCCCCAAAAA"]
```

Example 2:

```
Input: s = "AAAAAAAAAAAAA"
Output: ["AAAAAAAAAA"]
```

For example:

Input	Result
AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA

Answer: (penalty regime: 0 %)

```
s=input().strip()

seen=set()
repeated=[]

for i in range(len(s) - 9):
    sub = s[i:i+10]
    if sub in seen and sub not in repeated:
        repeated.append(sub)
    seen.add(sub)

for r in repeated:
    print(r)
```

	Input	Expected	Got	
✔	AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA	AAAAACCCCC CCCCCAAAAA	✔
✔	AAAAAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 7 | Correct Mark 1.00 out of 1.00

Write a program to eliminate the common elements in the given 2 arrays and print only the non-repeating

elements and the total number of such non-repeating elements.

Input Format:

The first line contains space-separated values, denoting the size of the two arrays in integer format respectively.

The next two lines contain the space-separated integer arrays to be compared.

Sample Input:

5 4

1 2 8 6 5

2 6 8 10

Sample Output:

1 5 10

3

Sample Input:

5 5

1 2 3 4 5

1 2 3 4 5

Sample Output:

NO SUCH ELEMENTS

For example:

Input	Result
5 4 1 2 8 6 5 2 6 8 10	1 5 10 3
5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS

Answer: (penalty regime: 0 %)

```
import sys

data=sys.stdin.read().split()
n1,n2=int(data[0]),int(data[1])
a=data[2:2+n1]
b=data[2+n1:2+n1+n2]

sa,sb=set(a),set(b)

result=[]
for x in a:
    if x not in sb and  x not in result:
        result.append(x)
for x in b:
    if x not in sa and x not in result:
        result.append(x)

if result:
    print(*result)
    print(len(result))
else:
    print('NO SUCH ELEMENTS')


```

	Input	Expected	Got	
✔	5 4 1 2 8 6 5 2 6 8 10	1 5 10 3	1 5 10 3	✔
✔	3 3 10 10 10 10 11 12	11 12 2	11 12 2	✔
✔	5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS	NO SUCH ELEMENTS	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 8 | Correct Mark 1.00 out of 1.00

Given an array of integers `nums` containing `n + 1` integers where each integer is in the range `[1, n]` inclusive.There is only **one repeated number** in `nums`, return *this repeated number*. Solve the problem using set.

Example 1:

```
Input: nums = [1,3,4,2,2]
```

```
Output: 2
```

Example 2:

```
Input: nums = [3,1,3,4,2]
```

```
Output: 3
```

For example:

Input	Result
1 3 4 4 2	4

Answer: (penalty regime: 0 %)

```
import sys

nums=sys.stdin.read().split()

seen=set()
for x in nums:
    if x in seen:
        print(x)
        break
    seen.add(x)
```

	Input	Expected	Got	
✔	1 3 4 4 2	4	4	✔
✔	1 2 2 3 4 5 6 7	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 9 | Correct Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
text=input().lower()

broken=set(input().strip().lower())

count=0
for word in text.split():
    if not(set(word) & broken):
        count +=1

print(count)
```

	Input	Expected	Got	
✔	hello world ad	1	1	✔
✔	Welcome to REC e	1	1	✔
✔	Faculty Upskilling in Python Programming ak	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 10 | Correct Mark 1.00 out of 1.00

Given a tuple and a positive integer k, the task is to find the count of distinct pairs in the tuple whose sum is equal to **K**.

Examples:

> **Input:** t = (5, 6, 5, 7, 7, 8), K = 13
> **Output:** 2
> **Explanation:**
> Pairs with sum K(= 13) are {(5, 8), (6, 7), (6, 7)}.
> Therefore, distinct pairs with sum K(= 13) are { (5, 8), (6, 7) }.
> Therefore, the required output is 2.

For example:

Input	Result
1,2,1,2,5 3	1
1,2 0	0

Answer: (penalty regime: 0 %)

```
t=tuple(int(x) for x in input().replace("(","").replace(")","").split(",") if x.strip())
k=int(input())

pairs=set()
for i in range(len(t)):
    for j in range(i+1,len(t)):
        if t[i] + t[j]  ==k:
            pairs.add((min(t[i], t[j]), max(t[i], t[j])))

print(len(pairs))
```

	Input	Expected	Got	
✔	5,6,5,7,7,8 13	2	2	✔
✔	1,2,1,2,5 3	1	1	✔
✔	1,2 0	0	0	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 7 October 2026, 11:45 PM
State	Finished
Completed on	Thursday, 8 October 2026, 12:50 PM
Time taken	13 hours 5 mins
Marks	5.00/5.00
Grade	**100.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
text=input().lower()
broken=set(input().strip().lower())

count=0
for work in text.split():
  if not (set(work) & broken):
     count +=1

print(count)
```

	Input	Expected	Got	
✔	hello world ad	1	1	✔
✔	Welcome to REC e	1	1	✔
✔	Faculty Upskilling in Python Programming ak	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

The program must accept a string S which represents a series of keystrokes as the input. There are two special keystrokes denoted by "^z" for undo operation and "^y" for redo operation. The undo operation removes the last typed character and redo operations reverses the immediately done last undo operations. The program must accept the keystrokes and perform all the operations (undo and redo). Finally, the program must print the string after applying all the operations as the output.

Boundary Condition(s):

1 <= Length of string <= 1000

Input Format:

The first line contains the string S.

Output Format:

The first line contains the modified string.

Example Input/Output 1:

Input:

missel^z^z^z^ypell^z^z^yt

Output:

misspelt

Explanation:

The characters **missel** are followed by three undo operations which deletes last three characters.

So the string **missel** becomes **mis**.

The three undo operations are followed by one redo operation which reverses the last undo operation.

So the string **mis** becomes **miss**.

Then the characters pell are entered which are followed by two undo operations so the last two characters are removed.

So the string becomes **misspe**.

Then a redo operation is applied which reverses the last undo operation and so the string **misspe** becomes **misspel**.

The characters are followed by **t** so the final string is misspelt.

Answer: (penalty regime: 0 %)

```
s=input()
result=[]
redo=[]
i=0
while i < len(s):
    if s[i]== '^' and i+1 < len(s) and s[i+1]== 'z':
        if result:
            redo.append(result.pop())
        i+=2
    elif s[i]== '^' and i+1 < len(s) and s[i+1] =='y':
```

```
        if redo:
            result.append(redo.pop())
        i+=2
    else:
        result.append(s[i])
        redo.clear()
        i+=1

print("".join(result))

```

	Input	Expected	Got	
✔	missel^z^z^z^ypell^z^z^yt	misspelt	misspelt	✔

Passed all tests! ✔

Correct

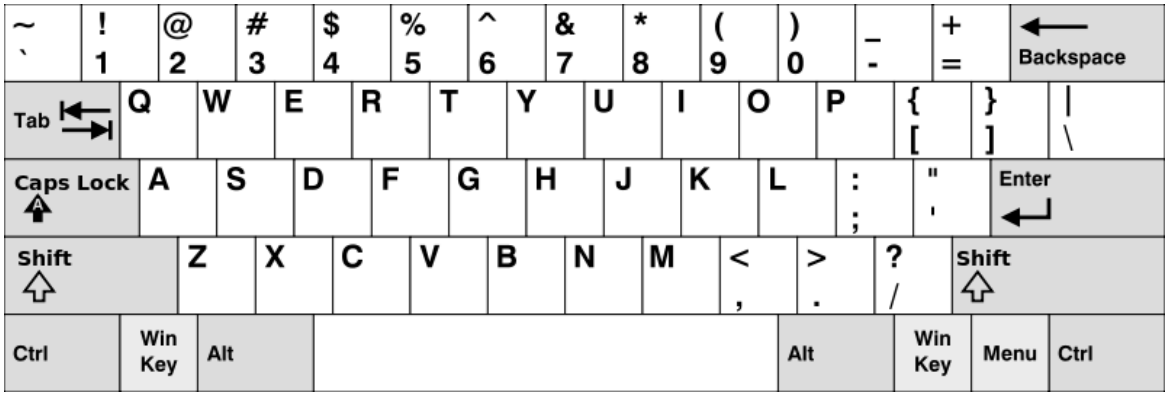
Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
n=int(input())
rows=[set("qweryuiop"),set("asdfghjkl"),set("zxcvbnm")]
found=False

for _ in range(n):
    word=input()
    letters=set(word.lower())
    if any(letters <= row for row in rows):
        print(word)
        found=True
```

```
if not found:
    print("No words")
```

	Input	Expected	Got	
✔	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✔
✔	1 omk	No words	No words	✔
✔	2 adsfd afd	adsfd afd	adsfd afd	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of lowercase and uppercase English letters.

A sentence can be shuffled by appending the 1-indexed word position to each word then rearranging the words in the sentence.

For example, the sentence "This is a sentence" can be shuffled as "sentence4 a3 is2 This1" or "is2 sentence4 This1 a3".

Given a shuffled sentence s containing no more than 9 words, reconstruct and return the original sentence.

Example 1:

Input:

is2 sentence4 This1 a3

Output:

This is a sentence

Explanation: Sort the words in s to their original positions "This1 is2 a3 sentence4", then remove the numbers.

Example 2:

Input:

Myself2 Me1 I4 and3

Output:

Me Myself and I

Explanation: Sort the words in s to their original positions "Me1 Myself2 and3 I4", then remove the numbers.

Constraints:

2 <= s.length <= 200

s consists of lowercase and uppercase English letters, spaces, and digits from 1 to 9.

The number of words in s is between 1 and 9.

The words in s are separated by a single space.

s contains no leading or trailing spaces.

Answer: (penalty regime: 0 %)

```
s=input()
words=s.split()
words.sort(key=lambda w: int(w[-1]))
print(" ".join(w[:-1] for w in words))
```

	Input	Expected	Got	
✔	`is2 sentence4 This1 a3`	`This is a sentence`	`This is a sentence`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

Given a number, convert it into corresponding alphabet.

Input	Output
1	A
26	Z
27	AA
676	YZ

Input Format

Input is an integer

Output Format

Print the alphabets

Constraints

1 <= num <= 4294967295

Sample Input 1

26

Sample Output 1

Z

For example:

Test	Result
`print(excelNumber(26))`	Z

Answer: (penalty regime: 0 %)

Reset answer

```
def excelNumber(n):
    result= ""
    while n>0:
        n -= 1
        result=chr(ord("A") + n%26)+result
        n//=26
    return result
```

	Test	Expected	Got	
✔	`print(excelNumber(26))`	Z	Z	✔
✔	`print(excelNumber(27))`	AA	AA	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Sunday, 11 October 2026, 11:45 PM
State	Finished
Completed on	Friday, 16 October 2026, 11:45 PM
Time taken	5 days
Marks	5.00/10.00
Grade	**50.00** out of 100.00

Question 1 | Correct Mark 1.00 out of 1.00

You are given a string `word`. A letter is called **special** if it appears both in lowercase and uppercase in `word`.

Your task is to return the number of **special** letters in `word`.

Constraints

- The input string `word` will contain only alphabetic characters (both lowercase and uppercase).
- The solution must utilize a dictionary to determine the number of special letters.
- The function should handle various edge cases, such as strings without any special letters, strings with only lowercase or uppercase letters, and mixed strings.

Examples

Example 1:

Input: word = "aaAbcBC"
Output: 3

Explanation:
The special characters in `word` are 'a', 'b', and 'c'.

Example 2:

Input: word = "abc"
Output: 0

Explanation:
No character in `word` appears in uppercase.

For example:

Test	Result
`print(count_special_letters("AaBbCcDdEe"))`	5

Answer: (penalty regime: 0 %)

Reset answer

```
def count_special_letters(word):
  d={}
  for ch in word:
    d[ch]=True
  count=0
  for ch in d:
    if ch.islower() and ch.upper() in d:
      count += 1
  return count
```

	Test	Expected	Got	
✔	`print(count_special_letters("AaBbCcDdEe"))`	5	5	✔

	Test	Expected	Got	
✔	`print(count_special_letters("ABCDE"))`	`0`	`0`	✔
✔	`print(count_special_letters("abcde"))`	`0`	`0`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Objective:

Develop a Python program that takes an input string from the user and counts the number of occurrences of each vowel (a, e, i, o, u) in the string. The program should be case-insensitive, meaning it should treat uppercase and lowercase vowels as the same.

Description:

Vowels play a significant role in the English language and other alphabet-based languages. Counting vowels in a given string is a fundamental task that can be applied in various text processing applications, including speech recognition, linguistic research, and text analysis. The objective of this problem is to create a Python script that accurately counts and displays the number of times each vowel appears in a user-provided string.

Program Requirements:

Input:

First line reading String as input, The string can contain any characters, including letters, numbers, and special characters.

Output:

Display the number of occurrences of each vowel in the string.

The output should list each vowel followed by its count.

Example:

Consider the following example for better understanding:

- **Input:** "Python Programming"
- **Output**

  a = 1
  e = 0
  i = 1
  o = 2
  u = 0

For example:

Input	Result
Hello World	a = 0 e = 1 i = 0 o = 2 u = 0
Python	a = 0 e = 0 i = 0 o = 1 u = 0

Answer: (penalty regime: 0 %)

```
s=input().lower()

vowels={'a':0,'e':0,'i':0,'o':0,'u':0}

for ch in s:
    if ch in vowels:
        vowels[ch] += 1

for v in vowels:
    print(v,"=", vowels[v])
```

	Input	Expected	Got	
✔	Hello World	a = 0 e = 1 i = 0 o = 2 u = 0	a = 0 e = 1 i = 0 o = 2 u = 0	✔
✔	AEIOU aeio u	a = 2 e = 2 i = 2 o = 2 u = 2	a = 2 e = 2 i = 2 o = 2 u = 2	✔
✔	Python	a = 0 e = 0 i = 0 o = 1 u = 0	a = 0 e = 0 i = 0 o = 1 u = 0	✔
✔	abcdefghijklmnopqrstuvwxyz	a = 1 e = 1 i = 1 o = 1 u = 1	a = 1 e = 1 i = 1 o = 1 u = 1	✔
✔	12345!@#$%AEIOU	a = 1 e = 1 i = 1 o = 1 u = 1	a = 1 e = 1 i = 1 o = 1 u = 1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 3 | Correct Mark 1.00 out of 1.00

A company wants to send its quotation secretly to its client. The company decided to encrypt the amount they are sending to their client with some special symbols so that the equation amount will not be revealed to any external person. They used the special symbols !,@,#,$,%,^,&,*,>,< for 0,1,2,3,4,5,6,7,8,9 respectively. Write a python code to help the company to convert the amount to special symbols.

(Value rounded off to 2 decimal points)

Input

n: Float data type which reads amount to send

Output

s: : String data type which displays symbols

Sample Testcase 1

Input

10000

Output

@!!!!.!!

Sample Testcase2

1234.56

Output

@#$%.^&

For example:

Input	Result
1345.23	@$%^.#$
15000.59	@^!!!.^<
156789	@^&*><.!!

Answer: (penalty regime: 0 %)

```
n=float(input())

symbols = {'0':'!','1':'@','2':'#','3':'$','4':'%','5':'^','6':'&','7':'*','8':'>','9':'<'}

s=f"{n:.2f}"
result=""
for ch in s:
    if ch in symbols:
        result += symbols[ch]
    else:
        result += ch

print(result)
```

	Input	Expected	Got	
✔	1345.23	@$%^.#$	@$%^.#$	✔

	Input	Expected	Got	
✔	15000.59	@^!!!.^<	@^!!!.^<	✔
✔	1234	@#$%.!!	@#$%.!!	✔
✔	156789	@^&*><.!!	@^&*><.!!	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 4 | Correct Mark 1.00 out of 1.00

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of lowercase and uppercase English letters.

A sentence can be shuffled by appending the 1-indexed word position to each word then rearranging the words in the sentence.

For example, the sentence "This is a sentence" can be shuffled as "sentence4 a3 is2 This1" or "is2 sentence4 This1 a3".

Given a shuffled sentence s containing no more than 9 words, reconstruct and return the original sentence.

Example 1:

Input:

is2 sentence4 This1 a3

Output:

This is a sentence

Explanation: Sort the words in s to their original positions "This1 is2 a3 sentence4", then remove the numbers.

Example 2:

Input:

Myself2 Me1 I4 and3

Output:

Me Myself and I

Explanation: Sort the words in s to their original positions "Me1 Myself2 and3 I4", then remove the numbers.

Constraints:

2 <= s.length <= 200

s consists of lowercase and uppercase English letters, spaces, and digits from 1 to 9.

The number of words in s is between 1 and 9.

The words in s are separated by a single space.

s contains no leading or trailing spaces.

Answer: (penalty regime: 0 %)

```
s=input()

d={}
for w in s.split():
    d[int(w[-1])]=w[:-1]

result=[]
for i in sorted(d):
    result.append(d[i])

print(" ".join(result))
```

	Input	Expected	Got	
✔	`is2 sentence4 This1 a3`	`This is a sentence`	`This is a sentence`	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 5 | Correct Mark 1.00 out of 1.00

A sentence is a string of single-space separated words where each word consists only of lowercase letters.A word is uncommon if it appears exactly once in one of the sentences, and does not appear in the other sentence.

Given two sentences s1 and s2, return a list of all the uncommon words. You may return the answer in any order.

Example 1:

Input: s1 = "this apple is sweet", s2 = "this apple is sour"

Output: ["sweet","sour"]

Example 2:

Input: s1 = "apple apple", s2 = "banana"

Output: ["banana"]

Constraints:

1 <= s1.length, s2.length <= 200

s1 and s2 consist of lowercase English letters and spaces.

s1 and s2 do not have leading or trailing spaces.

All the words in s1 and s2 are separated by a single space.

Note:

Use dictionary to solve the problem

For example:

Input	Result
this apple is sweet this apple is sour	sweet sour

Answer: (penalty regime: 0 %)

```
s1=input()
s2=input()

d={}
for w in (s1 + " " + s2 ).split():
    d[w]= d.get(w,0) +1

result=[]
for w in d:
    if d[w] == 1:
        result.append(w)

print(" ".join(result))
```

	Input	Expected	Got	
✔	this apple is sweet this apple is sour	sweet sour	sweet sour	✔

	Input	Expected	Got	
✔	apple apple banana	banana	banana	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 6 | Not answered Mark 0.00 out of 1.00

Create a student dictionary for n students with the student name as key and their test mark assignment mark and lab mark as values. Do the following computations and display the result.

1.Identify the student with the highest average score

2.Identify the student who as the highest Assignment marks

3.Identify the student with the Lowest lab marks

4.Identify the student with the lowest average score

Note:

If more than one student has the same score display all the student names

Sample input:

4

James 67 89 56

Lalith 89 45 45

Ram 89 89 89

Sita 70 70 70

Sample Output:

Ram

James Ram

Lalith

Lalith

For example:

Input	Result
4 James 67 89 56 Lalith 89 45 45 Ram 89 89 89 Sita 70 70 70	Ram James Ram Lalith Lalith

Answer: (penalty regime: 0 %)

```
1
```

Question 7 | Not answered Mark 0.00 out of 1.00

Given an array of names of candidates in an election. A candidate name in the array represents a vote cast to the candidate. Print the name of candidates received Max vote. If there is tie, print a lexicographically smaller name.

Examples:

Input : votes[] = {"john", "johnny", "jackie",

"johnny", "john", "jackie",

"jamie", "jamie", "john",

"johnny", "jamie", "johnny",

"john"};

Output : John

We have four Candidates with name as 'John', 'Johnny', 'jamie', 'jackie'. The candidates John and Johny get maximum votes. Since John is alphabetically smaller, we print it. Use dictionary to solve the above problem

Sample Input:

10

John

John

Johny

Jamie

Jamie

Johny

Jack

Johny

Johny

Jackie

Sample Output:

Johny

Answer: (penalty regime: 0 %)

```
1
```

Question 8 | Not answered Mark 0.00 out of 1.00

Given a number, convert it into corresponding alphabet.

Input	Output
1	A
26	Z
27	AA
676	YZ

Input Format

Input is an integer

Output Format

Print the alphabets

Constraints

1 <= num <= 4294967295

Sample Input 1

26

Sample Output 1

Z

For example:

Test	Result
`print(excelNumber(26))`	Z

Answer: (penalty regime: 0 %)

Reset answer

```
def excelNumber(n):
```

Question 9 | Not answered Mark 0.00 out of 1.00

Give a dictionary with value lists, sort the keys by summation of values in value list.

Input : test_dict = {'Gfg' : [6, 7, 4], 'best' : [7, 6, 5]}

Output : {'Gfg': 17, 'best': 18}

Explanation : Sorted by sum, and replaced.

Input : test_dict = {'Gfg' : [8,8], 'best' : [5,5]}

Output : {'best': 10, 'Gfg': 16}

Explanation : Sorted by sum, and replaced.

Sample Input:

2

Gfg 6 7 4

Best 7 6 5

Sample Output

Gfg 17

Best 18

For example:

Input	Result
2 Gfg 6 7 4 Best 7 6 5	Gfg 17 Best 18

Answer: (penalty regime: 0 %)

```
1
```

Question 10 | Not answered Mark 0.00 out of 1.00

In the game of Scrabble™, each letter has points associated with it. The total score of a word is the sum of the scores of its letters. More common letters are worth fewer points while less common letters are worth more points. The points associated with each letter are shown below:

Points Letters

1 A, E, I, L, N, O, R, S, T and U

2 D and G

3 B, C, M and P

4 F, H, V, W and Y

5 K

8 J and X

10 Q and Z

Write a program that computes and displays the Scrabble™ score for a word. Create a dictionary that maps from letters to point values. Then use the dictionary to compute the score.

A Scrabble™ board includes some squares that multiply the value of a letter or the value of an entire word. We will ignore these squares in this exercise.

Sample Input

REC

Sample Output

REC is worth 5 points.

For example:

Input	Result
REC	REC is worth 5 points.

Answer: (penalty regime: 0 %)

```
1
```